

UNIVERSITY COLLEGE OF ENGINEERING

End Semester Examination MATHEMATICS-IV (KAS302/KAS402)

Roll No. _____

Subject: MATHEMATICS-IV

Time: 3 Hours

Max. Marks: 84

Code: KAS302/KAS402

General Instructions:

1. Attempt all questions.
2. Calculators are allowed.
3. Assume any missing data suitably.

SECTION A — SECTION A

Attempt all questions. Each question carries 2 marks.

2 marks each | Attempt all questions

Q.No.	Question	Marks	CO
1	Form the partial differential equation by eliminating the arbitrary constants 'a' and 'b' from the relation $z = (x + a)(y + b)$.	2	CO1
2	Determine the nature of the second-order partial differential equation: $(x^2)u_{xx} + 2xyu_{xy} + (y^2)u_{yy} = 0$.	2	CO2
3	Calculate the first moment about the origin for the set of numbers $\{2, 4, 6, 8, 10\}$.	2	CO3
4	If a random variable X follows a Poisson distribution such that $P(X=1) = P(X=2)$, find the value of $P(X=0)$.	2	CO4
5	A continuous random variable X has a probability density function $f(x) = kx$ for $0 < x < 2$ and 0 otherwise. Determine the value of the constant k.	2	CO4
6	Find the standard error of the mean for a sample of size 400 drawn from an infinite population with a standard deviation of 15.	2	CO5
7	Calculate the degrees of freedom for a Chi-square test applied to a 4x3 contingency table.	2	CO5

SECTION B — SECTION B*Attempt all questions. Each question carries 7 marks.**7 marks each | Attempt all questions*

Q.No.	Question	Marks	CO
8	Solve the non-linear partial differential equation $p^2 + q^2 = x + y$ using Charpit's method.	7	CO1
9	Solve the one-dimensional heat equation $du/dt = c^2 (d^2u/dx^2)$ given the boundary conditions $u(0, t) = 0$, $u(L, t) = 0$ and the initial condition $u(x, 0) = x(L-x)$.	7	CO2
10	Fit a second-degree parabola of the form $y = a + bx + cx^2$ to the following data using the method of least squares: (x: 0, 1, 2, 3, 4) and (y: 1, 1.8, 1.3, 2.5, 6.3).	7	CO3
11	In a certain factory, machines A, B, and C produce 20%, 30%, and 50% of the total output respectively. Of their output, 5%, 4%, and 2% are defective. If an item is drawn at random and is found to be defective, compute the probability that it was produced by machine B.	7	CO4
12	A random sample of 10 individuals has the following IQs: 70, 120, 110, 101, 88, 83, 95, 98, 107, 100. Test the hypothesis that the population mean IQ is 100 at 5% level of significance. (Given t-table value for 9 d.f. at 5% is 2.262)	7	CO5

SECTION C — SECTION C

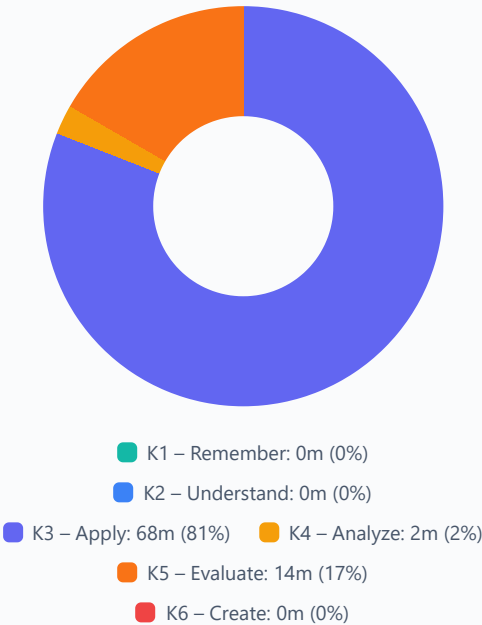
Attempt one part from each question. Each part carries 7 marks.

7 marks each | Attempt any one part of each question

Q.No.	Question	Marks	CO
13	(a) Solve the linear partial differential equation: $(x^2 - y^2 - z^2)p + 2xyq = 2xz$ using Lagrange's method.	7	CO1
	<i>OR</i>		
	(b) Find the complete solution of the higher-order PDE: $(D^2 - 4DD' + 4D'^2)z = e^{(2x+y)}$.	7	CO1
14	(a) A tightly stretched string of length L with fixed ends is initially at rest in its equilibrium position. If it is set vibrating by giving each point a velocity $v = kx(L-x)$, determine the displacement $u(x, t)$ at any time t .	7	CO2
	<i>OR</i>		
	(b) Solve the Laplace equation $d^2u/dx^2 + d^2u/dy^2 = 0$ in a rectangle $0 < x < a$, $0 < y < b$, given that $u(0, y) = 0$, $u(a, y) = 0$, $u(x, b) = 0$ and $u(x, 0) = f(x)$.	7	CO2
15	(a) Calculate the first four moments about the mean for the following frequency distribution: (x: 2, 3, 4, 5, 6) and (f: 1, 3, 7, 3, 1). Also compute the value of Kurtosis.	7	CO3
	<i>OR</i>		
	(b) Find the moment generating function of a random variable X having the density function $f(x) = e^{-x}$ for $x > 0$ and 0 otherwise. Use it to find the mean and variance.	7	CO3
16	(a) If X is a normally distributed random variable with mean 50 and standard deviation 10, find $P(X < 45)$ and $P(30 < X < 60)$. (Given Area under standard normal curve from 0 to 0.5 is 0.1915, from 0 to 1 is 0.3413, and from 0 to 2 is 0.4772)	7	CO4
	<i>OR</i>		
	(b) A discrete random variable X has the following probability distribution: (X: 0, 1, 2, 3) with probabilities (P: k , $2k$, $2k$, k^2). Find the value of k and calculate the expectation $E(2X + 3)$.	7	CO4
17	(a) In a sample of 1000 people, 540 are rice eaters and the rest are wheat eaters. Can we assume that both rice and wheat are equally popular in the population at 1% level of significance? (Critical value at 1% is 2.58)	7	CO5
	<i>OR</i>		
	(b) The following data shows the number of defectives in 10 samples of size 100 each: 12, 15, 13, 18, 10, 14, 16, 11, 19, 12. Construct a p-chart and determine if the process is under control.	7	CO5

Question Paper Analysis

Bloom's Taxonomy Distribution



Course Outcome Distribution

